Mihai Flavius Pirjoleanu

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Professional Summary

Full-Stack Developer & AI Researcher specializing in scalable system architectures and RAG-based AI implementations. Proven track record in optimizing enterprise workflows, from engineering high-uptime educational platforms to architecting low-latency microservices. Experience in bridging the gap between raw data science (NLP, CNNs) and production-ready software solutions with a focus on system reliability and performance.

Education

B.Sc. Cybernetics, Statistics and Computer Science | Academy of Economic Studies (ASE) **2023 – 2026**
Bucharest, Romania

- **Relevant Coursework:** Advanced Algorithms, Database Systems (Oracle/SQL), Software Design Patterns, Statistical Analysis, Data Structures.
- **Bachelor's Thesis:** AI-driven collaboration platform utilizing Vector Search and Semantic Matching, built with React, ASP.NET Core, and PostgreSQL.

Technical Skills

AI/ML:	TensorFlow, PyTorch, RAG Systems, Vector DBs (Chroma/Pinecone), CNNs, NLP, Scikit-learn
Programming:	Python, C# (.NET Core), JavaScript/TypeScript, C/C++, SQL, PL/SQL, R
Web Stack:	React.js, Next.js, Node.js/Express, FastAPI, RESTful API Design, HTML5/CSS3
Databases:	Oracle DB, PostgreSQL, MongoDB, Redis, ETL Pipeline Design
Data Science:	Pandas, NumPy, Multivariate Regression, ARIMA/Prophet Forecasting, PCA
Tools & DevOps:	Docker, Git, CI/CD Pipelines (GitHub Actions), Jest, Agile/Scrum, Unit Testing

Work Experience

Full-Stack Developer | ASE Bucharest **Nov 2025 – Present**
Bucharest, Romania

- Architected core full-stack modules for an enterprise educational ERP, supporting 4,800+ administrative users with 99.5% SLA adherence.
- Refactored legacy SQL procedures and optimized RESTful services, resulting in a 60% reduction in server-side execution time.

Mobile & Web Developer Intern | IT Smart Systems **Sep 2025 – Nov 2025**
Bucharest, Romania

- Decoupled legacy monolith into high-performance Microservices, reducing API latency by 40% for an environment serving 10K+ Daily Active Users (DAU).
- Deployed 15+ production-ready features with 95% test coverage; streamlined CI/CD pipelines to cut release deployment time by half.

Digital Transformation Intern | Romanian National Post **Jul 2025 – Sep 2025**
Bucharest, Romania

- Engineered an automated ETL framework in Python for sanitizing 10M+ geospatial records, utilizing fuzzy-matching algorithms to increase routing accuracy by 35%.
- Aggregated multi-source data from 5,000+ nodes, optimizing last-mile delivery logistics through Regex-based validation.

Web & AI Developer Intern | Devmind **Mar 2025 – Apr 2025**
Bucharest, Romania

- Deployed a production-ready RAG pipeline (GPT-4 + Vector DB) for 10,000+ technical documents, slashing query retrieval time by 95% (from 5m to 15s).
- Integrated LLM-driven chat interfaces, boosting platform user retention by 30% through improved semantic search accuracy.

Hackathons & Competitions

Top 10% – HacktheART ***2026***

- Solved 4+ advanced algorithmic problems involving abstract mathematics, number theory, and dynamic programming, maintaining a high-percentile ranking .

1st Place – FIICODE Amazon WebDev Hackathon ***2025***

- Led a 3-person team to build a MERN mobility platform with a custom AI route-matching algorithm, handling 300+ concurrent transactions with sub-200ms response times.

Research Top 10% – AI4Alzheimer Hack4Health ***2025***

- Developed a Hybrid CNN model for medical imaging, achieving 97.6% diagnostic accuracy on 130,000+ MRI scans, validated by healthcare professionals.

5th Place – Adobe WebDev Hackathon ***2024***

- Engineered an Offline-First React Native app with BLE mesh networking for P2P messaging, achieving 95% packet delivery rate without internet connectivity.

Research & Projects

Open-Tax Engine | Scalable ANAF & EU Compliance Suite ***2025 – Present***

- Developed an offline-first invoicing engine validating 16,000+ UBL permutations against ANAF/EU schemas, achieving a 0% validation error rate in production simulations.

NetPulse | High-Performance Custom Transport Protocol ***2026***

- Engineered a hybrid TCP/UDP communication layer in C/Python, achieving sub-3ms latency and 98.7% reliability over 1M+ packets under high-congestion stress tests.

LegisCast | AI-Driven Legislative Forecasting Platform ***2026***

- Architected a RAG-based system using vector search and ARIMA models to interpret economic policy, delivering impact forecasts with **92% predictive accuracy**.

Sameday Geo-Optimizer | Predictive Logistics Analytics ***2026***

- Built a Streamlit geo-analytics engine for 6,000+ lockers, processing 250k deliveries to boost demand prediction by 18% via K-Means expansion clustering.

EuroStat Insight | Econometric Life Expectancy Modeling ***2025***

- Designed a multivariate regression framework for EU-wide socio-economic data, identifying mortality predictors with a 96% statistical confidence level.

GlobalHealth PCA | Education-Health Correlation Analysis ***2025***

- Applied PCA and OLS modeling on data from 112 countries to demonstrate that education metrics explain 89% of global health outcome variance.

Volunteering & Leadership

Python Mentor | Societe Generale "Dalia's Book" **Oct 2024 – May 2025**

Remote

- Orchestrated weekly Python workshops for 20+ students, designing a custom curriculum covering OOP principles and game development.
- Achieved an 85% course completion rate, mentoring students in building their first functional software applications.

Certifications

UiPath Process Mining Developer	2025
Oracle Database@AWS Architect	2025
Oracle PL/SQL Developer	2025
Oracle SQL Developer	2024

Languages

Romanian: Native **English:** C1 (CAE Cambridge) – Professional Proficiency